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Convertible pet stroller and carrier

Abstract

An assembly comprising a pet stroller with a removably attached pet carrier. The pet carrier has front and rear access to the enclosed space and is useable separately from the stroller.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1623259	12/1926	McGregor et al.	N/A	N/A
2365003	12/1943	Reinholz	N/A	N/A
3116069	12/1962	Dostal	N/A	N/A
3159850	12/1963	Aldrich et al.	N/A	N/A
3349413	12/1966	Merlis	N/A	N/A
4412688	12/1982	Giordani	280/658	B62B 7/08
4481685	12/1983	Watson	N/A	N/A
4707024	12/1986	Schrader	N/A	N/A
D308656	12/1989	Takahashi	D12/129	N/A
5074575	12/1990	Bigo	280/647	B62B 7/10
5201535	12/1992	Kato	297/229	B62B 7/123
5203577	12/1992	Kato	280/47.38	B62B 7/142
D344701	12/1993	Church	D12/129	N/A
5479892	12/1995	Edwards	N/A	N/A
5524965	12/1995	Barley	N/A	N/A
5553885	12/1995	Chang	280/658	B62B 7/14
5772279	12/1997	Johnson, Jr.	N/A	N/A
D398083	12/1997	Martz	N/A	N/A
D432464	12/1999	Gehr	N/A	N/A
6139046	12/1999	Aalund	280/47.38	B62B 9/20
6223691	12/2000	Beattie	N/A	N/A
D445966	12/2000	Licciardello	N/A	N/A
6267404	12/2000	Yang et al.	N/A	N/A
D453593	12/2001	Licciardello	N/A	N/A
D455680	12/2001	Huang	N/A	N/A
6416076	12/2001	Hou et al.	N/A	N/A
D463329	12/2001	Yang et al.	N/A	N/A
D469929	12/2002	Licciardello	N/A	N/A
D475660	12/2002	Licciardello	N/A	N/A
6755435	12/2003	Hsia	280/47.38	B62B 7/083
6766930	12/2003	Dixon et al.	N/A	N/A
6910709	12/2004	Chen	N/A	N/A
D510549	12/2004	Kassai et al.	N/A	N/A
7011318	12/2005	Chen	N/A	N/A
7090242	12/2005	Sheinall et al.	N/A	N/A
D569049	12/2007	Lin	N/A	N/A
D574757	12/2007	Chen	N/A	N/A
7500692	12/2008	Espenshade	N/A	N/A

7568450	12/2008	Chen	N/A	N/A
D608505	12/2009	Lin	N/A	N/A
7665421	12/2009	Martz	N/A	N/A
7753397	12/2009	Yang	N/A	N/A
7862069	12/2010	King	N/A	N/A
D654406	12/2011	Henry	N/A	N/A
8186705	12/2011	Greger et al.	N/A	N/A
8360498	12/2012	Ostad et al.	N/A	N/A
8651502	12/2013	Winterhalter	280/643	B60N 2/2848
8708364	12/2013	Gower et al.	N/A	N/A
D707896	12/2013	Renforth	N/A	N/A
8955856	12/2014	Guo	N/A	N/A
D723746	12/2014	Jakubowski	N/A	N/A
9108659	12/2014	Sparling	N/A	B62B 7/142
D757370	12/2015	Jakubowski et al.	N/A	N/A
D793914	12/2016	Eriksson	N/A	N/A
D801592	12/2016	Jakubowski et al.	N/A	N/A
D802489	12/2016	Stiba	N/A	N/A
D808084	12/2017	Lu	N/A	N/A
D815567	12/2017	Oakes	N/A	N/A
D827939	12/2017	Jakubowski et al.	N/A	N/A
D833918	12/2017	Gopaul	N/A	N/A
D855899	12/2018	Jakubowski et al.	N/A	N/A
D856602	12/2018	Lu	N/A	N/A
D869099	12/2018	Lu	N/A	N/A
D869344	12/2018	Oakes	N/A	N/A
D874987	12/2019	Baker et al.	N/A	N/A
D875619	12/2019	Conley	N/A	N/A
D878249	12/2019	Pos	N/A	N/A
D882184	12/2019	Jakubowski et al.	N/A	N/A
D883860	12/2019	Barenbrug	N/A	N/A
D886535	12/2019	Zemel	N/A	N/A
10667489	12/2019	Jakubowski et al.	N/A	N/A
D889323	12/2019	Reinius et al.	N/A	N/A
D891993	12/2019	Altenhofen et al.	N/A	N/A
D893359	12/2019	Holleis et al.	N/A	N/A
D897909	12/2019	Xiang	N/A	N/A
D899006	12/2019	Chen	N/A	N/A
D906900	12/2020	Walker et al.	N/A	N/A
D915247	12/2020	Haley et al.	N/A	N/A
D919488	12/2020	Napier	N/A	N/A
D920588	12/2020	Chen	N/A	N/A
D923256	12/2020	Jakubowski	N/A	N/A
D926639	12/2020	Hauser et al.	N/A	N/A
D930262	12/2020	Chaudhry et al.	N/A	N/A
D950425	12/2021	Lin	N/A	N/A
D955060	12/2021	Wu	N/A	N/A
D955301	12/2021	Lin	N/A	N/A
D958703	12/2021	Adler	N/A	N/A

D960770	12/2021	Jakubowski	N/A	N/A
D969033	12/2021	Huntley	N/A	N/A
D982838	12/2022	Han	N/A	N/A
D984931	12/2022	Bowers	N/A	N/A
D988611	12/2022	Pang	N/A	N/A
D989663	12/2022	Van Gelderen	N/A	N/A
D991104	12/2022	Huntley	N/A	N/A
D991576	12/2022	Yu	N/A	N/A
D993542	12/2022	Pang	N/A	N/A
D996732	12/2022	Jakubowski	N/A	N/A
D996733	12/2022	Yu	N/A	N/A
D997466	12/2022	Tu	N/A	N/A
D999681	12/2022	Adler	N/A	N/A
D1000001	12/2022	Lu	N/A	N/A
11760403	12/2022	Yoo	N/A	N/A
D1000717	12/2022	Lu	N/A	N/A
D1002443	12/2022	Holper	N/A	N/A
D1003530	12/2022	Tu	N/A	N/A
D1003531	12/2022	Tu	N/A	N/A
D1004209	12/2022	Chen	N/A	N/A
D1007770	12/2022	Pang	N/A	N/A
D1012377	12/2023	Cal	N/A	N/A
D1026741	12/2023	Jakubowski	N/A	N/A
11993301	12/2023	Cheng	N/A	N/A
11993302	12/2023	Zheng	N/A	N/A
D1032419	12/2023	Fang	N/A	N/A
D1035167	12/2023	Jakubowski	N/A	N/A
D1039449	12/2023	Yi	N/A	N/A
D1039452	12/2023	Fang	N/A	N/A
12060101	12/2023	Su	N/A	N/A
D1042243	12/2023	Kho	N/A	N/A
D1045239	12/2023	Jakubowski	N/A	N/A
D1045683	12/2023	Yi	N/A	N/A
D1047781	12/2023	Gopaul	N/A	N/A
2005/0168006	12/2004	Darland	N/A	N/A
2006/0169218	12/2005	Chang	N/A	N/A
2008/0012408	12/2007	Santamaria	N/A	N/A
2011/0278810	12/2010	Winterhalter et al.	N/A	N/A
2013/0113185	12/2012	Zehfuss	280/647	B62B 7/14
2014/0159358	12/2013	Chen	280/769	B62B 9/12
2014/0230747	12/2013	Jakubowski	N/A	N/A
2014/0333106	12/2013	Chen	N/A	N/A
2015/0076774	12/2014	Sclare	280/33.993	B62B 7/145
2016/0021843	12/2015	Jakubowski	N/A	N/A
2016/0157458	12/2015	Jakubowski	N/A	N/A
2016/0229439	12/2015	Zhong	N/A	N/A
2016/0339940	12/2015	Lee et al.	N/A	N/A
2018/0132446	12/2017	Jakubowski	N/A	N/A
2019/0002007	12/2018	Xiang	N/A	N/A

2019/0053462	12/2018	Lin	N/A	N/A
2019/0133073	12/2018	Lu	N/A	N/A
2020/0236895	12/2019	Cheung	N/A	A01K 1/0272
2020/0247453	12/2019	Gibson et al.	N/A	N/A
2020/0296919	12/2019	Jakubowski	N/A	N/A
2020/0353967	12/2019	Horst et al.	N/A	N/A
2021/0022314	12/2020	Xiang	N/A	N/A
2021/0127627	12/2020	Jakubowski	N/A	N/A
2022/0297737	12/2021	Horst et al.	N/A	N/A
2023/0159078	12/2022	Zhu	N/A	N/A
2023/0397570	12/2022	Cheng	N/A	N/A
2023/0415807	12/2022	Wu	N/A	N/A
2024/0017759	12/2023	Sack	N/A	N/A
2024/0067251	12/2023	Zhu	N/A	N/A
2024/0067252	12/2023	Zhong	N/A	N/A
2024/0075976	12/2023	Jablonski	N/A	N/A
2024/0092412	12/2023	Zhong	N/A	N/A
2024/0246592	12/2023	Gower	N/A	N/A
2024/0246593	12/2023	Saucier	N/A	N/A
2024/0253684	12/2023	Huang	N/A	N/A
2024/0278820	12/2023	Horst	N/A	N/A
2024/0326892	12/2023	Zehfuss	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201258003	12/2008	CN	N/A
100584657	12/2009	CN	B60N 2/2806
102190012	12/2010	CN	N/A
103407466	12/2012	CN	N/A
203876796	12/2013	CN	N/A
205327136	12/2015	CN	N/A
105882721	12/2015	CN	N/A
205632628	12/2015	CN	N/A
206005460	12/2016	CN	N/A
207201652	12/2017	CN	N/A
208113692	12/2017	CN	N/A
208149362	12/2017	CN	N/A
109050643	12/2017	CN	B62B 7/12
213168254	12/2020	CN	N/A
2268220	12/1993	GB	B62B 7/14
2511200	12/2013	GB	N/A
2006254801	12/2005	JP	N/A
3133057	12/2006	JP	N/A
2007185180	12/2006	JP	N/A
2009247278	12/2008	JP	N/A
2013123413	12/2012	JP	N/A
2015107700	12/2014	JP	N/A
2018023324	12/2017	JP	N/A

OTHER PUBLICATIONS

CN-207201652-U FIT database Translation (Year: 2018). cited by examiner
JP-2018023324-A FIT database Translation (Year: 2018). cited by examiner
JP-2015107700-A FIT database Translation (Year: 2015). cited by examiner
JP-2013123413-A FIT database Translation (Year: 2013). cited by examiner
JP-2009247278-A FIT Database Translation (Year: 2009). cited by examiner
JP-2007185180-A FIT Database Translation (Year: 2007). cited by examiner
JP-3133057-U FIT Database Translation (Year: 2007). cited by examiner
JP-2006254801-A FIT Database Translation (Year: 2006). cited by examiner
CN208149362U and Translation (Year: 2017). cited by examiner
GB-2268220-A Translation (Year: 1994). cited by examiner
CN-109050643-A Translation (Year: 2018). cited by examiner
CN-100584657-C Translation (Year: 2010). cited by examiner
CN-201258003-Y Translation (Year: 2009). cited by examiner
CN-213168254-U Translation (Year: 2021). cited by examiner
Utopian pet four wheel pet stroller, <https://lots-of-pets.myshopify.com/products/utopian-pet-four-wheel-pet-stroller-color-olive>. cited by applicant
IBIYAYA elegant retro I, <https://www.amazon.com/ibiyaya-FS1202-NB-Elegant-Retro-Stroller/dp/B00M1OAUEE>. cited by applicant
Paws & Pals 4 Wheel 2-in-1 Stroller & Carrier, <https://www.chewy.com/paws-pals-2-in-1-detachable-dog-cat/dp/217526>. cited by applicant
Aosom Elite-Jr Dog Pet Bike Trailer / Stroller with Swivel Wheel—Red / Black, <https://www.aosom.com/item/aosom-elite-jr-dog-pet-bike-trailer-stroller-w-swivel-wheel-red-black~5663-1334R-B.html>. cited by applicant
HPZ PetRover USA, 3-in-1 Pet Travel System, New 2019 Model, 6 pgs. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/823,544, filed Mar. 19, 2020, which claims the benefit of U.S. Patent Application No. 62/820,594, filed Mar. 19, 2019, which are incorporated by reference as if fully set forth.

FIELD

(1) The disclosure relates to pet carriers, car seat frames, pet strollers, and assemblies of pet carriers with car seat frames or pet strollers.

BACKGROUND

(2) Many people enjoy taking their pets with them to a variety of locations. Often the pet may be carried. In such a case, a pet carrier may be useful. Devices to carry pets and transport pets within carriers are desired.

SUMMARY

(3) In an aspect, the invention relates to a pet carrier comprising a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section. The top section comprises a

front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back canopy latch element at the back. The canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section. A first canopy frame element of the at least one canopy frame element extends at least partially around the lower edge of the canopy and has a front section at the front and a back section at the back. The front section and the back section meet at a first pivot on a first side of the canopy and meet at a second pivot on a second side of the canopy. The front canopy latch element is fixed to the front section and the back canopy latch element is fixed to the back section. The bottom section comprises a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back. The shell extends up from the base to the bottom frame element. The bottom frame element extends around the upper rim of the bottom, and the front and the back bottom latch elements are fixed to the bottom frame element. The handle comprises a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier. The handle also comprises a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier. The handle also comprises a handle bar extending from the first handle receiver to the second handle receiver and over the canopy. The front canopy latch element is engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position, and is releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier. The back canopy latch element is engageable with the back bottom latch element to secure the back of the top to the back of the bottom in a back closed position, and is releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier. The first pet carrier engagement is on the first side of the bottom section. The second pet carrier engagement is on the second side of the bottom section. Both the first pet carrier engagement and the second pet carrier engagement are engageable with a secondary device in order to fix the pet carrier to the secondary device. The top section, the bottom section, and the base form an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

(4) In an aspect, the invention relates to a car seat frame. The car set frame comprises a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end. The first side comprises a first side carrier support, a first vertical support on the first side, and a second vertical support on the first side. Each vertical support on the first side having a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support. The second side comprising a second side carrier support, a first vertical support on the second side, and a second vertical support on the second side. Each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support. The second end of the first vertical support on the first side is fixed to the first end of the first base element. The second end of the second vertical support on the first side is fixed to the first end of the second base element. The second end of the first vertical support on the second side is fixed to the second end of the first base element. The second end of the second vertical support on the second side is fixed to the second end of the second based element. The car seat frame further comprises a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support.

(5) In an aspect, the invention relates to an assembly comprising a pet carrier and the car seat frame as summarized above.

(6) In an aspect the invention relates to a pet stroller comprising a frame and wheels. The frame comprises a first side, a second side, a push bar, a carrier support. Each side comprises a front lower portion, a rear lower portion, a mid portion, and an upper portion. Each of the front and rear

lower portions having first end and a wheel engagement at the first end to which one of the wheels is engaged. The front and rear lower portions on the first side each have a second end opposite the respective first ends that converge at a first junction. The front and rear lower portions on the second side each have a second end opposite the respective first end that converges at a second junction. The mid portion on the first side has a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion on the second side has a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion on the first side has a first end extending from the second end of the mid portion on the first side and a second end. The upper portion on the second side has a first end extending from the second end of the mid portion on the second side and a second end. The push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the second end of the upper portion on the second side. The carrier support is configured to receive a pet carrier.

(7) In an aspect, the invention relates to an assembly comprising a pet carrier and the pet stroller summarized above.

(8) In an aspect, the invention relates to a system comprising at least two of the above summarized pet carrier, car seat frame, and pet stroller.

(9) In an aspect, the invention relates to a pet carrier. The pet carrier comprises a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section. The top section comprises a front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back canopy latch element at the back. The canopy comprises a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a back section at the back. The front section and the back section meet at a first pivot on a first side of the canopy. The front section and the back section also meet at a second pivot on a second side of the canopy. The front canopy latch element is fixed to the front section, and the back canopy latch element is fixed to the back section. The bottom section comprises a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back. The shell extends up from the base to the bottom frame element. The bottom frame element extends around the upper rim of the bottom. The front and the back bottom latch elements are fixed to the bottom frame element. The handle comprising a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier, and a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier. The handle also comprises a handle bar extending from the first handle receiver to the second handle receiver and over the canopy. The front canopy latch element being engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position. The front canopy latch element also being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier. The back canopy latch element being engageable with the back bottom latch element to secure the back of the top to the back of the bottom in a back closed position. The back canopy latch element also being releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier. The first pet carrier engagement on the first side of the bottom section. The second pet carrier engagement on the second side of the bottom section. Both the first pet carrier engagement and the second pet carrier engagement being engageable with respective receivers on a secondary device in order to fix the pet carrier to the secondary device. The top section, the bottom section, and the base forming an enclosure, in which a pet may be

contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

(10) In another aspect, the invention relates to a car seat frame. The car seat frame comprises a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end. The first side comprises a first side carrier support, a first vertical support on the first side, and a second vertical support on the first side. Each vertical support on the first side has a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support. The second side comprises a second side carrier support, a first vertical support on the second side, and a second vertical support on the second side. Each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support. The second end of the first vertical support on the first side is fixed to the first end of the first base element. The second end of the second vertical support on the first side is fixed to the first end of the second base element. The second end of the first vertical support on the second side is fixed to the second end of the first base element, and the second end of the second vertical support on the second side is fixed to the second end of the second base element. The car seat frame further comprises a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support. The car seat frame may comprise one or more seat belt retainers, where each seat belt retainer is fixed to one of the vertical supports. The one or more seat belt retainers may comprise a first seat belt retainer fixed to the first vertical support on the first side, and a second seat belt retainer fixed to the second vertical support on the first side. The car seat frame may comprise a first frame-carrier receiver fixed to the first side carrier support and a second frame-carrier receiver fixed to the second side carrier support.

(11) In another aspect, the invention relates to an assembly comprising a pet carrier and the car seat frame. The pet carrier may be the pet carrier summarized above. When the pet carrier is the pet carrier summarized above, the first pet carrier engagement may be engaged with the first frame-carrier receiver, and the second pet carrier engagement may be engaged with the second frame-carrier receiver.

(12) In an aspect, the invention relates to a pet stroller comprising a frame and four wheels. The frame comprises a first side, a second side, a push bar, a pet carrier support. Each side comprises a front lower portion, a rear lower portion, a mid portion, and an upper portion. Each of the front and rear lower portions having a first end and a wheel engagement at the first end to which one of the wheels is engaged. The front and rear lower portions on the first side each have a second end opposite the respective first ends that converge at a first junction. The front and rear lower portions on the second side each have a second end opposite the respective first end that converge at a second junction. The mid portion on the first side has a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion on the second side has a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion on the first side has a first end extending from the second end of the mid portion on the first side and a second end. The upper portion on the second side has a first end extending from the second end of the mid portion on the second side and a second end. The push bar has a first end connected to the second end of the upper portion on the first side, and the second end connected to the second end of the upper portion on the second side.

(13) In another aspect, the invention relates to an assembly comprising a pet carrier and the pet stroller summarized above. The pet carrier may rest upon the front and rear carrier supports. The pet carrier may be the pet carrier summarized above. When the pet carrier is the pet carrier summarized above, the first and second pet carrier engagements may be respectively engaged with the first and second pet carrier receivers.

(14) In an aspect, the invention relates to a system comprising at least two of the above summarized aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following detailed description of embodiments of the present invention will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there are shown in the drawings embodiments which are presently preferred. It is understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:
- (2) FIG. 1 illustrates a perspective view of a pet carrier.
- (3) FIG. 2 illustrates a first side view of the pet carrier.
- (4) FIG. 3 illustrates a second side view of the pet carrier.
- (5) FIG. 4 illustrates a bottom view of the pet carrier.
- (6) FIG. 5 illustrates a car seat frame.
- (7) FIG. 6 illustrates a pet stroller.
- (8) FIG. 7 illustrates a side view of the pet stroller.
- (9) FIG. 8 illustrates a bottom view of the pet stroller.
- (10) FIG. 9 illustrates an assembly of a pet carrier and a car seat frame.
- (11) FIG. 10 illustrates an assembly of a pet carrier and a pet stroller.
- (12) FIG. 11 illustrates a car seat frame exemplifying alternate frame-carrier receivers and alternate seat belt retainers.
- (13) FIG. 12 illustrates a pet carrier with pet carrier engagements in combination with the car seat frame of FIG. 11.
- (14) FIG. 13 illustrates one of the pet carrier engagements on the pet carrier illustrated in FIG. 12 engaged with one of the frame-carrier receivers of the car seat frame of FIG. 11.
- (15) FIG. 14 illustrates one of the seat belt retainers on the car seat frame of FIG. 11.
- (16) FIG. 15 illustrates a pet carrier having pet carrier engagements in combination with a stroller.
- (17) FIG. 16 illustrates the combination of FIG. 15 with one of the pet carrier engagements in an open position but on the stroller.
- (18) FIG. 17 illustrates a perspective view of the pet carrier and the stroller of FIG. 15 in an un-combined state.
- (19) FIG. 18 illustrates a front view of the pet carrier and the stroller of FIG. 15 in an un-combined state.
- (20) FIG. 19 illustrates a side view of the pet carrier and the stroller of FIG. 15 in an un-combined state.
- (21) FIG. 20 illustrates a rear view of the pet carrier and the stroller of FIG. 15 in an un-combined state.

DETAILED DESCRIPTION

- (22) Certain terminology is used in the following description for convenience only and is not limiting. The words “right,” “left,” “top,” and “bottom” designate directions in the drawings to which reference is made.
- (23) The words “a” and “one,” as used in the claims and in the corresponding portions of the specification, are defined as including one or more of the referenced item unless specifically stated otherwise. The phrase “at least one” followed by a list of two or more items, such as “A, B, or C,” means any individual one of A, B or C as well as any combination thereof.
- (24) An embodiment includes a pet carrier. Referring to FIGS. 1-4, an embodiment is illustrated as pet carrier **100**, which may comprise a top section **102**, a bottom section **104** having an upper rim

125, a handle **106**, a first pet carrier engagement **120a**, a second pet carrier engagement **120b**, and an interior defined between the top section and the bottom section.

(25) The top section **102** may comprise a front, a back, and a canopy **108**. The top section **102** may also comprise at least one canopy frame element supporting the canopy, a front canopy latch element **110a** at the front, and a back canopy latch element **110b** at the back, as illustrated for top section **102**. Ridges **111a**, **111b**, **111c**, and **111d** may be present, and may be seams of different panels of the canopy, and/or engagement areas between the canopy at frame elements inside the canopy. Frame elements inside the canopy **108** may be enclosed within or tethered to canopy material. The canopy **108** may also comprise a lower edge **109** configured to be proximal to the upper rim **125** of the bottom section **104**. A first canopy frame element **111** of the at least one canopy frame element having a front section **126** and a back section **127** may extend at least partially around the lower edge of the canopy. The front section **126** and the back section **127** may meet at a first pivot **128** on a first side of the canopy. The front section **126** and the back section **127** may also meet at a second pivot **129** on a second side of the canopy. The front section **126**, back section **127**, first pivot **128**, and second pivot configured such that one or both of the front and back sides of the canopy may be rotated upward or downward. The front canopy latch element **110a** may be attached to the front section **126** of the first canopy frame element **111**, and the back canopy latch element **110b** may be fixed to the back section **127** of the first canopy frame element **111**. The frame elements other than the first canopy frame element **111** may each comprise a first end associated with an attachment point on the first side of the pet carrier, and a second end associated with an attachment point on the second side of the pet carrier. The attachment points may be at the first pivot **128** and the second pivot **129** such that the one or both of the front and back sides of the canopy may be rotated upward or downward. Intermediate portions, between the first and second ends, of these frame elements may be configured to contact, be contained within the material of, or be tethered to the canopy **108**.

(26) The pet carrier **100** has the bottom section **104**, which may comprise a base **124**, a front, a back, a first side, a second side, a shell **105**, at least one bottom frame element, a front bottom latch element **115a** on the front, and a back bottom latch element **115b** on the back. A first bottom frame element **114** of the at least one bottom frame element may extend around the upper rim **125** of the bottom section **104**. The front and the back bottom latch elements (**115a**, **115b**) may be fixed to the first bottom frame element **114**.

(27) The material of a portion or the whole of at least one of the canopy or the shell may be opaque, translucent, solid, mesh. Any type of material may be used. The material may be nylon.

(28) A pet carrier may include a handle, which may be the handle **106** illustrated. The handle **106** may comprise a handle bar **130**, a first handle receiver **107a**, and a second handle receiver **107b**. The first handle receiver **107a** may be engaged with at least one of the first canopy frame element **111** or the first bottom frame element **114** on the first side of the pet carrier. The second handle receiver **107b** may be engaged with at least one of the first canopy frame element **111** or the first bottom frame element **115b** on the second side of the pet carrier. The handle bar **130** may extend from the first handle receiver **107a** to the second handle receiver **107b** and over the canopy **108**. Each handle receiver may be separate from the handle bar **130** and affixed to it, or integral with the handle bar **130**. Each handle receiver may also be affixed to at least one of the first canopy frame element **111** or the first bottom frame element **114** or integral thereto. In an embodiment, the handle **106** is fixed in position. In an embodiment, each handle receiver may comprise a handle release (**107c**, **107d**), which when activated allow rotation of the handle bar **130** over the top of the pet carrier **100**.

(29) The handle **106** may be comprised of more than one part or be a single piece. The pet carrier may have a grip **131**, that is either a separate piece affixed to the handle bar **130** or integral with the handle bar **130**.

(30) The front canopy latch element **110a** may be engageable with the front bottom latch element

115a to secure the front of the top section **102** to the front of the bottom section **104** in a front closed position. The front canopy latch element **110a** may also be releasable to an open position, allowing movement of the top section **102** at the front of the pet carrier to reveal the interior of the pet carrier.

(31) The back canopy latch element **110b** may be engageable with the back bottom latch element **115b** to secure the back of the top section **102** to the back of the bottom section **104** in a back closed position. The back canopy latch element **110b** may be capable of being released to an open position, allowing movement of the top section **102** at the back of the canopy to reveal the interior of the pet carrier.

(32) The first pet carrier engagement **120a** is illustrated on the first side of the bottom section **104**, and extending toward the base **124**. The second pet carrier engagement **120b** is illustrated on the second side of the bottom section **104**, and extending toward the base **124**. Both the first pet carrier engagement **120a** and the second pet carrier engagement **120b** may be capable of engaging respective receivers on a secondary device in order to fix the pet carrier to the secondary device. The secondary device may be a car seat frame or a pet stroller. The carrier engagements **120a**, **120b** may be affixed to or integral with the first bottom frame element **114**, or other structures on the pet carrier. In an embodiment, the carrier engagements **120a**, **120b** are at alternate locations on the pet carrier to fit to a specific secondary device. In an embodiment, the carrier engagements **120a**, **120b** may have alternate structures/mechanisms and/or be moved forward or backward along the pet carrier to so that the pet carrier may be fit to various secondary devices.

(33) The top section **102** and the bottom section **104** may form an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position. A pet may also be resident within the bottom section **104** while at least one or both of the front and back portions of the top section **102** are raised. The pet may then be able to view the surrounding environment, or others may be able to interact with the pet visually or by touch.

(34) The pet carrier illustrated further comprises a first support wire **116** proximal to the front of the bottom section **104**. The first support wire may be engaged with the first bottom frame element **114** on the first and second sides of the bottom section **104**, and is illustrated extending from the first bottom frame element **114** along the first side of the bottom section **104** toward the base **124**, across the base **124**, and along the second side of the bottom section **104** toward the first bottom frame element **114**. The pet carrier illustrated also further comprises a second support wire **118** proximal to the rear of the bottom section **104**, engaged with the first bottom frame element **114** on the first and second sides of the bottom section **104**, and is illustrated extending from the first bottom frame element **114** along the first side of the bottom section **104** toward the base **124**, across the base **124**, and along the second side of the bottom section **104** toward the first bottom frame element **114**. The embodiment illustrated shows the support wires exterior to the shell **105**. One or both of the first support wire **116** and second support wire **118** may be positioned exterior to the bottom section **104**, interior to the bottom section **104**, or within the material of the bottom section **104**. The support wires **116**, **118** may stabilize the structure of the pet carrier **100**. The support wires **116**, **118** may be utilized to stabilize the bottom when the pet carrier **100** is set on a surface. The embodiment illustrated includes wire holders **210a**, **210b**, **310a**, and **310b** that secure the support wires to the base **124**, and may stabilize the positioning of the support wires. Embodiments of pet carriers herein may include alternate support structures to stabilize the pet carrier.

(35) An embodiment includes a car seat frame. Referring to FIG. 5, an embodiment is illustrated as car seat frame **500**, which may comprise a first side, a second side, a first base element **514** having a first end and a second end, and a second base element **516** having a first end and a second end.

(36) The first side may comprise a first side carrier support **502**, a first vertical support on the first side **506**, and a second vertical support on the first side **508**. Each of vertical supports **506**, **508** have a first end and a second end, and the first end of each is fixed to opposite ends of the first side

carrier support **502**.

(37) The second side may comprise a second side carrier support **504**, a first vertical support on the second side **510**, and a second vertical support on the second side **512**. Each of vertical supports **510**, **512** have a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support **504**.

(38) As illustrated, the second end of the first vertical support on the first side **506** is fixed to the first end of the first base element **514**, and the second end of the second vertical support on the first side **508** is fixed to the first end of the second base element **516**. The second end of the first vertical support on the second side **510** is fixed to the second end of the first base element **514**, and the second end of the second vertical support on the second side **512** is fixed to the second end of the second based element **516**.

(39) FIG. 5 illustrates an embodiment of the car seat frame **500** where the first vertical supports **506**, **510** on the first and second sides are integral with the first base element **514**, and the first vertical supports **506**, **510** are integral with a portion of the first side carrier support **502** and second side carrier support **504**, respectively. This portion of the car seat frame may be referred to as a first piece. The first side carrier support **502** and second side carrier support **504** each include a junction where a non-integral second piece is fixed and extends to the second end of each of the first side carrier support **502** and second side carrier support **504**. The first piece ends at the junction, and the remaining portion of first side carrier support **502** and second side carrier support **504** in combination with the second vertical supports **508**, **512** and second base element **516** may be referred to as the second piece of the car seat frame **500**. These elements may be constructed as illustrated, or by any other arrangement. They may all be integral with one another, or divided into two or separate pieces that are fixed together by any suitable fastener or junction element. The material of these elements may be any material sufficient to create the structure, and withstand the anticipated weight of a pet carrier and pet set on the car seat frame. For example, the material may be metal or polymeric. The material may be hollow or solid. The material may be hollow tubing.

(40) A car seat frame may comprise one or more seat belt retainers. Seat belt retainers **522a** and **522b** are illustrated in FIG. 5 and may be used to secure the car seat frame **500** on a car seat with the seat belt fitted with the retainers. Seat belt retainers **524a**, **524b** are optional for the FIG. 5 embodiment, and if present would allow securing the car seat frame **500** to the seat in the opposite direction. Seat belt retainers may be integral with or separate from the vertical supports.

(41) A car set may be secured to a car seat frame in embodiments herein in many different fashions. For example, a car seat could be strapped to a car seat frame. The car seat frame **500** comprises a first frame-carrier receiver **518** fixed to the first side carrier support **502** and a second frame-carrier receiver **520** fixed to the second side carrier support **504**. The frame-carrier receivers may engage matching engagements of a carrier. For example, the frame-carrier receiver **518** and frame-carrier receiver **520** may engage pet carrier engagement **120a** and pet carrier engagement **120b** of pet carrier **100** (see FIGS. 1-4).

(42) An embodiment includes an assembly comprising a pet carrier and a car seat frame, where the pet carrier is secured to the car seat frame. Optionally, the car seat frame is also secured to a seat. The seat may be the seat of a car. The car seat frame could be secured to the seat of the car by wrapping a seat belt of the seat around elements of the frame. An embodiment of the assembly includes the pet carrier **100** secured to the car seat frame **500**, as illustrated in FIG. 9. As illustrated in FIG. 9, the frame-carrier receiver **518** and frame-carrier receiver **520** engage pet carrier engagement **120a** and pet carrier engagement **120b** of pet carrier **100**. The assembly of pet carrier **100** and car seat frame **500** may further comprise a seat comprising a seat belt, where the seat belt is arranged in the seat belt retainer **522a** and seat belt retainer **522b**, or the optional seat belt retainer **524a** and seat belt retainer **524b**, to secure the car seat frame **500** to the seat. The seat may be a car seat.

(43) An embodiment includes a pet stroller that may carry a pet carrier. The pet stroller may include

wheels in any configuration and supports on which a pet carrier may rest. It may also include restraints to secure the pet carrier to the pet stroller. Referring to FIGS. 6-8, an embodiment is illustrated as pet stroller **600**, which comprises a frame **601** and four wheels (wheel **658**, wheel **660**, wheel **662**, and wheel **664**).

(44) As illustrated, the frame comprising a first side **602**, a second side **604**, a push bar **631**, a front pet carrier support **624**, and a rear pet carrier support **628**. Each of the first side **602** and the second side **604** comprises a front lower portion (**610**, **616**), a rear lower portion (**611**, **620**), a mid-portion (**614**, **615**), and an upper portion (**612**, **613**). Each of the front and rear lower portions have a first end and a wheel engagement at the first end to which one of the wheels is engaged. The front **610** and rear **611** lower portions on the first side each having a second end opposite the respective first ends that converge at a first junction **621a**. The front **616** and rear **620** lower portions on the second side each have a second end opposite the respective first end that converges at a second junction **621b**. The mid portion **614** on the first side has a first end at the first junction **621a** and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion **615** on the second side has a first end at the second junction **621b** and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion **612** on the first side is illustrated with a first end extending from the second end of the mid portion **614** on the first side and a second end, the upper portion **613** on the second side having a first end extending from the second end of the mid portion **615** on the second side and a second end. The push bar **631** is illustrated with a first end connected to the second end of the upper portion **612** on the first side, and a second end connected to the second end of the upper portion **613** on the second side. The specific arrangement illustrated may be varied in embodiments herein. For example, other angles between the above mention elements are within the scope of embodiments herein.

(45) The frame may be made of separate elements fixed to one another. Alternatively, one or more of the elements may be integral with one another. The material of the frame may be any that supports the utility of the pet stroller. The frame may be comprised completely or in part by elements that are polymeric or metal. The frame may be comprised completely or in part by elements that are either hollow or solid. Any suitable fasteners may be used to fasten separate frame elements to one another. For example, fastener may be chosen from at least one of rivets, nuts and bolts, screws, junction elements made to fit and secure the various frame elements, etc. This type of material and construction may apply to all other elements of the pet stroller **800**. For example, the front pet carrier support **624**, the rear pet carrier support **628**, the first intermediate cross member **626**, and rear cross member **810** may be fashioned in a similar manner. Connectors **634**, **636**, **638**, **640**, **642**, **644**, **646**, and **648** are illustrated in FIGS. 6-8. A connector may connect separate pieces of a pet stroller. Alternatively, a connector may be positioned on a single element to provide attachment points for other elements.

(46) The front pet carrier support **624** is illustrated extending between and fixed to the front lower portions **610**, **616**. Similarly, the rear pet carrier support **628** is illustrated extending between and fixed to the rear lower portions **611**, **620**.

(47) A pet stroller may simply provide a space for a pet carrier, and optionally supports for a pet stroller. A user then could chose to secure the pet carrier to the pet stroller in any suitable fashion. For example, a pet carrier could be strapped to elements of the pet stroller. The pet stroller embodiment of FIGS. 6-8 comprise first stroller-carrier receiver **674** fixed to the first junction **621a** and facing the second junction **621b**. The pet stroller of FIGS. 6-8 also comprises a second stroller-carrier receiver **676** fixed to the second junction **621b** and facing the first junction **621a**. The first stroller-carrier receiver **674** and the second stroller-carrier receiver are capable of engaging pet carrier engagements in order to fix a pet carrier to the pet stroller.

(48) An embodiment includes an assembly comprising a pet carrier and a pet stroller, where the pet

carrier is received in or secured to the pet stroller. The pet carrier may be secured to the pet stroller by any suitable securements. For example, the pet carrier could be lashed to elements of the pet stroller. An embodiment of the assembly comprises the pet carrier **100** secured to the pet stroller **600**. See FIG. **10**. As illustrated in FIG. **10**, the first and second pet carrier engagements **120a**, **120b** are respectively engaged with the first and second stroller-carrier receivers **674**, **676**.

(49) An embodiment includes a system comprising a pet carrier, a car seat frame, and a pet stroller. The car seat frame and the pet stroller may be capable of receiving the pet carrier for transport of the pet carrier on either a car seat or the pet stroller. Any suitable method of receiving and/or securing the pet carrier to either the car seat frame or the pet stroller may be part of the system. The pet carrier may comprise one or more engagements that match one or more receivers on one or both of the car seat carrier and the pet stroller. The pet carrier may include one engagement type that matches receiver(s) on the car seat carrier, and a second engagement type that matches receiver(s) on the pet stroller. The engagement(s) of the pet carrier may be of one type, and the receivers on both the car seat carrier and the pet stroller may be configured to match the one type of engagement(s) on the pet carrier. The system may comprise the pet carrier **100**, the car seat frame **500**, and the pet stroller **600**. As illustrated, the first stroller-carrier receiver **674**, the second stroller-carrier receiver **676**, the first frame-carrier receiver **518**, and the second frame-carrier receiver **520** are illustrated as matching such that the pet carrier **100** may be assembled with either the car seat frame **500** or the pet stroller **600**.

(50) As stated above, the carrier engagements **120a**, **120b** may have alternate structures/mechanisms and/or be moved forward or backward along the pet carrier so that the pet carrier can fit to various secondary devices. The carrier engagements herein are non-limiting examples of carrier engagements or carrier engagement means of embodiments herein. Likewise, the carrier receivers and carrier supports on secondary devices may have alternate structures/mechanisms and/or be moved forward or backward along their respective elements to accommodate alternate carrier engagements. The receivers and carrier supports herein are non-limiting examples of receivers or receiver means and carriers or carrier means of embodiments herein. Pet carriers, car seat frames, and strollers may have alternate accessory structures/mechanisms as well. Non-limiting examples include a car seat frame that may have alternate seat belt retainers, a car seat frame that may have an alternate configuration of supports positioning the frame-carrier receivers at the correct height, or a stroller with front and back carrier supports contiguous with one another to encircle a carrier.

(51) Referring to FIG. **11**, an embodiment of a car seat frame **1101** exemplifying alternate frame-carrier receivers and alternate seat belt retainers is illustrated. A frame-carrier receiver **1118**, on a first side carrier support **1102**, is illustrated opposed to a frame-carrier receiver **1120**, on a second side carrier support **1104**. The frame-carrier receiver comprises a retainer **1118a** and a retainer **1118b**. In the embodiment illustrated, the retainer **1118a** includes an inclined surface **1119a**, and the retainer **1118b** includes an inclined surface **1119b**. The two inclined surfaces **1119a** and **1119b** are angled toward each other. In this configuration, a pet carrier engagement of a carrier being positioned upon the car seat frame may be guided into the frame carrier receiver **1118**. The retainers may also engage with structures on a pet carrier to aid in position and/or retention of the pet carrier. A frame carrier receiver **1120** is illustrated opposed to the frame-carrier receiver **1118**. Like the frame-carrier-receiver **1118**, the frame-carrier receiver **1120** includes retainers **1120a** and **1120b**, with inclined surfaces **1121a** and **1121b**.

(52) FIGS. **11** and **14** also illustrate alternate seat belt retainers. One or more of seat belt retainers **1122a**, **1122b**, **1124a**, and **1124b** may be included on a car seat frame herein. Referring to seat belt retainer **1122a**, the retainer may include an elongated body comprising a slot **1401** and a slot **1403**. The retainer may also include a protrusion **1405** and a protrusion **1407** on a vertical support of the car seat frame. The protrusion **1405** and the protrusion **1407** may be spaced such that a seat belt may be arranged between them. The protrusion **1405** and the protrusion **1407** may also be

spaced such that the slot **1401** and the slot **1403** of the elongated body may be respectively fitted thereon. As illustrated, the slots **1401**, **1401** may be configured to slide from a widened opening that receives a larger end of the protrusion **1405**, **1407**, respectively, down to a narrowed opening over a narrowed portion of the protrusion **1405**, **1407**, thereby locking the elongated body onto the protrusions. A seat belt may be aligned between the protrusion **1405** and **1407** prior to engaging the elongated body, and in such a manner lock the seat belt to the car seat frame. The seat belt retainer **1122a** also includes a user control tab **1409**, which may be engaged by a user to operate the seat belt retainer **1122a**. One seat belt retainer is illustrated in FIG. **14**. One or more seat belt retainer of a car seat frame may be similarly configured.

(53) FIG. **12** illustrates a pet carrier **1299** with a pet carrier engagement **1201** in combination with the car seat frame of FIG. **11**. As illustrated, the pet carrier engagement **1201** is positioned between the retainers **1118a** and **1118b**.

(54) FIG. **13** illustrates the pet carrier engagement **1201** on the pet carrier illustrated in FIG. **12** engaged with the frame-carrier receiver, like that of frame-carrier receiver **1120** in FIG. **11**. The pet carrier engagement **1201** includes a mount **1301**, which is fixed to the bottom section **1304** of a pet carrier **1299**. A mount may, as illustrated for mount **1301**, be fixed to the a first bottom frame element **1314**. A pet carrier engagement may also include, as illustrated for pet carrier engagement **1201**, a rotating latch **1380**. The rotating latch may be connected to a mount. In the embodiment illustrated, the rotating latch **1380** includes pins received in holes within the mount **1301**. One pin **1381** is illustrated in one hole **1382**. A similar pin and hole to match is on the far side of the rotating latch **1380**/mount **1301** combination (not shown). The rotating latch **1380** includes an opening (see FIG. **16**) that may be fitted around the second side carrier support **1104**. The open position of rotating latch **1380** in FIG. **16** is shown in relation to a stroller. However, a similar configuration may be envisioned between the rotating latch **1380** and the second side carrier support **1104**. Once the rotating latch is in position over the support it may be rotated to close upon the support. In FIG. **13**, the rotating latch **1380** is illustrated in a position where it has been fitted around the second side carrier support **1104**. It may be rotated further downward to a closed position. It may lock in the closed position. The lock may be any suitable lock. The lock may be press fitted engagements that may be released by flexing the rotating latch **1380** or forcing the engagements past one another. A pet carrier may include a pet carrier engagement like pet carrier engagement **1201** on each side, in place of pet carrier engagements **120a** and **120b** of FIGS. **1** and **2**.

(55) FIGS. **15-20** illustrate the pet carrier **1299** having alternate pet carrier engagements and a stroller **1501**. The pet carrier engagement **1201** is illustrated on one side of the pet carrier **1299**, and a pet carrier engagement **1801** is illustrated on the other side of the pet carrier engagement **1201**.

(56) Referring to FIG. **17**, the stroller **1501** includes a carrier support **1503**. The carrier support **1503** may include a carrier support frame **1505** including a first side support **1510** connected to the first side of the stroller and having a front end **1511** and a rear end **1512**, and a second side support **1520** connected to the second side of the stroller and having a front end **1521** and rear end **1522**. The carrier support may also include a third side **1530** extending from the front end **1511** of the first side support **1510** and the front end **1521** of the second side support **1520**, and a fourth side **1540** extending from the rear end **1512** of the first side support **1510** and the rear end **1522** of the second side support **1520**. A stroller herein may be configured as known to the skilled artisan to present a wheeled and pushable support system for a pet carrier. The basic elements may be similar to those of the stroller of FIG. **6**. Alternatively, a carrier support **1503**, as illustrated in FIGS. **15-20**, may replace the front carrier support **624** and the rear carrier support **628**. The two different carrier support configurations for the strollers of FIG. **6** versus FIGS. **15-20** represent non-limiting means by which a stroller may support a carrier. Generally, a stroller may comprise a carrier support. The carrier support may be two or more different structures. See the non-limiting example of FIG. **6**. The carrier support may be a single support (which may comprise a single integral unit, or be

comprise of parts engaged together to form the single support). See the non-limiting example of FIGS. 15-20. The support(s) represent non-limiting examples of a support(s) or support means of embodiments herein.

(57) FIG. 15 illustrates the pet carrier **1299** in combination with the stroller **1501**. The pet carrier includes the pet carrier engagement **1201**, but in FIG. 15 it is engaged on the carrier support **1503** of a stroller **1501**. The engagement of the carrier support **1503** by the pet carrier engagement **1201** is similar to that described above for the car set frame in FIGS. 11-13. FIG. 15 illustrates the pet carrier engagement **1201** closed upon the carrier support **1503**. FIG. 16 illustrates the pet carrier engagement **1201** fitted onto the carrier support **1503** but in the open position.

(58) FIGS. 17-20 illustrate different views of the stroller **1501** and the pet carrier **1299**. The carrier support **1503** may prevent forward or reward movement of the pet carrier **1299**. The carrier support may include first and second stroller-carrier receivers. The first and second stroller-carrier receivers may be, but are not necessarily limited to ones similar or the same in configuration and operation as the frame carrier receivers **1118** and **1120**, or the stroller-carrier receivers **674**, **676**.

(59) As stated above, an embodiment includes an assembly comprising a pet carrier and a pet stroller, where the pet carrier is received in or secured to the pet stroller. The pet carrier may be secured to the pet stroller by any suitable securements. For example, the pet carrier could be lashed to elements of the pet stroller. An embodiment of the assembly comprises the pet carrier **1299** secured to the pet stroller **1501**.

(60) An embodiment includes an assembly comprising a pet carrier and a car seat frame, where the pet carrier is received in or secured to the car seat frame. The pet carrier may be secured to the car seat frame by any suitable securements. For example, the pet carrier could be lashed to elements of the car seat frame. An embodiment of the assembly comprises the pet carrier **1299** secured to the car seat frame **500**. An embodiment of the assembly comprises the pet carrier **100** secured to the car seat frame **1101**.

(61) An embodiment includes a system comprising a pet carrier, a car seat frame, and a pet stroller. The car seat frame and the pet stroller may be capable of receiving the pet carrier for transport of the pet carrier on either a car seat or the pet stroller. Any suitable method of receiving and/or securing the pet carrier to either the car seat frame or the pet stroller may be part of the system. The pet carrier may comprise one or more engagements that match one or more receivers on one or both of the car seat carrier and the pet stroller. The pet carrier may include one engagement type that matches receiver(s) on the car seat carrier, and a second engagement type that matches receiver(s) on the pet stroller. The engagement(s) of the pet carrier may be of one type, and the receivers on both the car seat carrier and the pet stroller may be configured to match the one type of engagement(s) on the pet carrier. The system may comprise the pet carrier **100**, the car seat frame **500**, and the pet stroller **600**. As illustrated, the first stroller-carrier receiver **674**, the second stroller-carrier receiver **676**, the first frame-carrier receiver **518**, and the second frame-carrier receiver **520** are illustrated as matching such that the pet carrier **100** may be assembled with either the car seat frame **500** or the pet stroller **600**. The system may comprise the pet carrier **1299**, the car seat frame **1101**, and the pet stroller **1501**. As illustrated, the pet carrier engagements **1201** and **1801** can be used to secure the pet carrier **1299** to either the car seat frame **1101** or the pet stroller **1501**.

(62) The following list of embodiments is a non-exclusive gathering of embodiments herein. Embodiments otherwise described herein are not excluded from the scope of what may be claimed by this list.

LIST OF EMBODIMENTS

(63) 1. A pet carrier comprising a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section; the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back

canopy latch element at the back, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a back section at the back, the front section and the back section meeting at a first pivot on a first side of the canopy and meeting at a second pivot on a second side of the canopy, the front canopy latch element fixed to the front section and the back canopy latch element fixed to the back section; the bottom section comprising a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back, the shell extends up from the base to the bottom frame element, the bottom frame element extends around the upper rim of the bottom, and the front and the back bottom latch elements are fixed to the bottom frame element; the handle comprising a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier, and a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier, a handle bar extending from the first handle receiver to the second handle receiver and over the canopy; the front canopy latch element being engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position, and being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier; the back canopy latch element being engageable with the back bottom latch element to secure the back of the top to the back of the bottom in a back closed position, and being releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier; the first pet carrier engagement on the first side of the bottom section, the second pet carrier engagement on the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with a secondary device in order to fix the pet carrier to the secondary device; and the top section, the bottom section, and the base forming an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

(64) 2. The pet carrier of embodiment 1 further comprising a first support wire proximal to the front of the bottom section, engaged with the first bottom frame element on first and second sides of the bottom section, and extending from the first bottom frame element along the first side of the bottom toward the base, across the base, and along the second side of the bottom toward the first bottom frame element, and a second support wire proximal to a rear of the bottom section, engaged with the first bottom frame element on the first and second sides of the bottom, and extending from the first bottom frame element along the first side of the bottom toward the base, across the base, and along the second side of the bottom toward the first bottom frame element, one or both of the first support wire and second support wire may be positioned either exterior to the bottom or interior to the bottom.

(65) 3. The pet carrier of embodiment 1 or embodiment 2, wherein the first pet carrier engagement extends toward the base, and the second pet carrier engagement extends toward the base.

(66) 4. The pet carrier of embodiment 1 or embodiment 2, wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, and a first rotating latch rotatably fixed to the first mount and configured to engage the secondary device, and the second pet carrier engagement comprises a second mount fixed to the bottom section, and a second rotating latch rotatably fixed to the second mount and configured to engage the secondary device.

(67) 5. A car seat frame comprising a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end; the first side comprising a first side carrier support, a first vertical support on the first side, and a second vertical support on the first side, each vertical support on the first side having a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support; the second side comprising a second side carrier support, a first vertical support on the second side, and a second

vertical support on the second side, each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support; and the second end of the first vertical support on the first side is fixed to the first end of the first base element, the second end of the second vertical support on the first side is fixed to the first end of the second base element, the second end of the first vertical support on the second side is fixed to the second end of the first base element, and the second end of the second vertical support on the second side is fixed to the second end of the second based element; the car seat frame further comprising a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support.

(68) 6. The car seat frame of embodiment 5, wherein the first frame-carrier receiver comprises a first retainer and a second retainer fixed on the first side carrier support, and a first side carrier support length of the first carrier support positioned between the first retainer and the second retainer, the first retainer and the second retainer spaced to receive a first pet carrier engagement therebetween, the second frame-carrier receiver comprises a third retainer and a fourth retainer fixed on the second side carrier support, and a second side carrier support length of the second side carrier support positioned between the third retainer and the fourth retainer, the third retainer and the fourth retainer spaced to receive a second pet carrier engagement therebetween.

(69) 7. The car seat frame embodiment 5 or embodiment 6 comprising one or more a seat belt retainers, where each seat belt retainer is fixed to one of the vertical supports.

(70) 8. The car seat frame of embodiment 7, wherein the one or more seat belt retainers comprise a first seat belt retainer fixed to the first vertical support on the first side and a second seat belt retainer fixed to the second vertical support on the first side.

(71) 9. An assembly comprising a pet carrier and the car seat frame of any one of embodiments 5-8.

(72) 10. The assembly of embodiment 9, wherein the pet carrier is the pet carrier of one of embodiments 1-4.

(73) 11. The assembly of embodiment 10, wherein the first pet carrier engagement is engaged with the first frame-carrier receiver, and the second pet carrier engagement is engaged with the second frame-carrier receiver.

(74) 12. A pet stroller comprising a frame and four wheels; the frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper portion; each of the front and rear lower portions having first end and a wheel engagement at the first end to which one of the wheels is engaged; the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge at a first junction, the front and rear lower portions on the second side each having a second end opposite the respective first end that converges at a second junction; the mid portion on the first side having a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction, the mid portion on the second side having a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction; the upper portion on the first side having a first end extending from the second end of the mid portion on the first side and a second end, the upper portion on the second side having a first end extending from the second end of the mid portion on the second side and a second end; the push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the second end of the upper portion on the second side; wherein the carrier support is configured to receive a pet carrier.

(75) 13. The pet stroller of embodiment 12, wherein the carrier support comprises a frame defining a rectangular opening into which a pet carrier may be received, a first side of the frame connected to the first side of the stroller and a second side of the frame connected to the second side of the stroller, and wherein the pet carrier engagements of a pet carrier may be received by the carrier support.

- (76) 14. The pet stroller of embodiment 13, wherein the first side of the frame is connected to the mid portion on the first side of the stroller and a second side of the frame is connected to the mid portion on the second side of the stroller.
- (77) 15. The pet stroller of embodiment 12, wherein the pet stroller further comprises a first stroller-carrier receiver fixed to the first junction and facing the second junction, and a second stroller-carrier receiver fixed to the second junction and facing the first junction, the first and second stroller-carrier receivers capable of engaging carrier engagements in order to fix a pet carrier to the pet stroller.
- (78) 16. The pet stroller of embodiment 12 or embodiment 15, wherein the carrier support comprises a front pet carrier support extending between and fixed to the front lower portions, and a rear pet carrier support extending between and fixed to the rear lower portions.
- (79) 17. An assembly comprising a pet carrier and the pet stroller of one of embodiments 12-16, wherein the pet carrier engagements engage the pet stroller.
- (80) 18. The assembly of embodiment 17, wherein the pet carrier is the pet carrier of one of embodiments 1-4.
- (81) 19. The assembly of embodiment 18, wherein the first and second pet carrier engagements are respectively engaged with either the carrier support or the first and second stroller-carrier receivers.
- (82) 20. A system comprising at least two of the pet carrier of one of embodiments 1-4, the car seat frame of one or more of embodiments 5-8, the assembly of one or more of embodiments 9-11, the pet stroller of one of embodiments 12-16, and the assembly of one of embodiments 17-19.
- (83) Further embodiments may be achieved by changing a structure in an embodiment herein to a corresponding structure in another embodiment herein. For example, the pet carrier engagements of FIGS. 1-3 may be used in systems herein in combination with car seats or strollers with receivers like FIGS. 4-10 but fitted onto a support like that of support 1503 of FIGS. 15-20.

REFERENCE CHARACTER LISTING

- (84) **100**—Pet Carrier **102**—Top Section **104**—Bottom Section **105**—Shell **106**—Handle **107a**—First Handle Receiver **107b**—Second Handle Receiver **107c**—First Handle Release **107d**—Second Handle Release **108**—Canopy **109**—Lower Edge **110a**—Front Canopy Latch Element **110b**—Rear Canopy Latch Element **111**—First Canopy Frame Element **111a**—Ridge **111b**—Ridge **111c**—Ridge **111d**—Ridge **112**—Carrier Grip **114**—First Bottom Frame Element **115a**—Front Bottom Latch Element **115b**—Back Bottom Latch Element **116**—First Support Wire **118**—Second Support Wire **120a**—Pet Carrier Engagement **120b**—Pet Carrier Engagement **124**—Base **125**—Upper Rim **126**—Front Section of First Canopy Frame Element **127**—Rear Section of First Canopy Frame Element **128**—First Pivot **129**—Second Pivot **130**—Handle Bar **131**—Grip **210a**—Wire Holder **210b**—Wire Holder **310a**—Wire Holder **310b**—Wire Holder **500**—Car Seat Frame **502**—First Side Carrier Support **504**—Second Side Carrier Support **506**—First Vertical Support on the First Side **508**—Second Vertical Support on the First Side **510**—First Vertical Support on the Second Side **512**—Second Vertical Support on the Second Side **514**—First Base Element **516**—Second Base Element **518**—Frame-Carrier Receiver **520**—Frame-Carrier Receiver **522a**—Seat Belt Retainer **522b**—Seat Belt Retainer **524a**—Seat Belt Retainer **524b**—Seat Belt Retainer **600**—Pet stroller **601**—Frame **602**—First Side **604**—Second Side **610**—Front First Side Lower Portion **611**—Rear First Side Lower Portion **612**—Upper Portion **613**—Upper Portion **614**—Mid Portion **615**—Mid Portion **616**—Front Second Side Lower Portion **620**—Rear Second Side Lower Portion **621a**—First Junction **621b**—Second Junction **622**—Brace **624**—Front Pet Carrier Support **625**—Brace **626**—First Intermediate Cross Member **628**—Rear Pet Carrier Support **630**—Grip **631**—Push Bar **632**—Release **634**—Connector **636**—Connector **638**—Connector **640**—Connector **642**—Connector **644**—Connector **646**—Connector **648**—Connector **650**—Front Panel **652**—First Side Panel **654**—Second Side Panel **656**—Bottom Panel **658**—Wheel **660**—Wheel **662**—Wheel **664**—Wheel **666**—Wheel Housing **668**—Wheel Housing **670**—Front Wheel Bearing Support **672**—Front Wheel Bearing Support **674**—First Stroller-Carrier Receiver **676**—Second Stroller-Carrier

Receiver **810**—Rear Cross Member **1101**—Car Seat Frame **1102**—First Side Carrier Support **1104**—Second Side Carrier Support **1118**—Frame-Carrier Receiver **1118a**—Retainer **1118b**—Retainer **1119a**—Inclined Surface **1119b**—Inclined Surface **1120**—Frame-Carrier Receiver **1120a**—Retainer **1120b**—Retainer **1121a**—Inclined Surface **1121b**—Inclined Surface **1201**—Pet Carrier Engagement **1201 1299**—Pet Carrier **1301**—Mount **1304**—Bottom Section **1314**—First Bottom Frame Element **1380**—Rotating Latch **1501**—Stroller **1503**—Carrier Support **1801**—Pet Carrier Engagement

(85) It is understood, therefore, that the invention is not limited to the particular embodiments disclosed, but is intended to cover all modifications which are within the spirit and scope of the invention as defined by the appended claims, the above description, and/or shown in the attached drawings.

Claims

1. An assembly comprising a pet stroller with a removably attached pet carrier: the pet stroller comprising: a stroller frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper portion; each of the front and rear lower portions having a first end and a wheel engagement at the first end to which a wheel is engaged; the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge toward one another, the front and rear lower portions on the second side each having a second end opposite the respective first end that converge toward one another; and the pet carrier comprising: a top section, a bottom section having an upper rim, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section; the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a pivotal connection to the upper rim; the bottom section comprising a base, a front, a back, a first side, a second side, a shell, and a bottom frame element, the shell extends up from the base to the bottom frame element, the bottom frame element extends around and defines the upper rim of the bottom; at least one of the top section at the front of the pet carrier or the top section at the back of the pet carrier is openable to reveal the interior of the pet carrier; the first pet carrier engagement at the first side of the bottom section, the second pet carrier engagement at the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with the carrier support of the stroller; and the top section, the bottom section, and the base forming an enclosure, in which a pet may be contained within the interior of the pet carrier when the front and the back of the canopy are in a closed position; wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, the first mount including hooked protrusions extending at either end of the first mount and a first rotating latch rotatably coupled to the first mount between the hooked protrusions, the hooked protrusions of the first mount and the first rotating latch are configured to engage a first side support of the carrier support of the stroller.

2. The assembly of claim 1, wherein the second pet carrier engagement comprises a second mount fixed to the bottom section, the second mount including hooked protrusions extending at either end of the second mount and a second rotating latch rotatably coupled to the second mount between the hooked protrusions, the hooked protrusions of the second mount and the second rotating latch are configured to engage a second side support of the carrier support of the stroller.

3. An assembly comprising a pet stroller with a removably attached pet carrier: the pet stroller comprising: a stroller frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper

portion; each of the front and rear lower portions having a first end and a wheel engagement at the first end to which a wheel is engaged; the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge toward one another, the front and rear lower portions on the second side each having a second end opposite the respective first end that converge toward one another; each mid portion being a singular bar having a first end connected to the second end of the respective front lower portion and a second end that extends in an opposite direction; the upper portion on the first side having a first end extending from the second end of the mid portion on the first side and a second end, the upper portion on the second side having a first end extending from the second end of the mid portion on the second side and a second end; the push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the second end of the upper portion on the second side; and a first brace extending from the rear lower portion on the first side to the upper portion on the first side and a second brace extending from the rear lower portion on the second side to the upper portion on the second side wherein the carrier support is located in a region of the mid portions and comprises a carrier support frame including a first side support connected only directly to the first side mid portion and the first brace and having a front end and a rear end, and a second side support connected only directly to the second side mid portion and the second brace and having a front end and rear end, a third side extending from the front end of the first side support to the front end of the second side support, and a fourth side extending from the rear end of the first side support to the rear end of the second side support, the carrier support configured to receive the pet carrier within the carrier support frame; and the pet carrier comprising: a top section, a bottom section having an upper rim, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section; the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a pivotal connection to the upper rim; the bottom section comprising a base, a front, a back, a first side, a second side, a shell, and a bottom frame element, the shell extends up from the base to the bottom frame element, the bottom frame element extends around and defines the upper rim of the bottom; at least one of the top section at the front of the pet carrier or the top section at the back of the pet carrier is openable to reveal the interior of the pet carrier; the first pet carrier engagement at the first side of the bottom section, the second pet carrier engagement at the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with the carrier support of the stroller; and the top section, the bottom section, and the base forming an enclosure, in which a pet may be contained within the interior of the pet carrier when the front and the back of the canopy are in the closed position; wherein the bottom section further comprises a front canopy latch on the front, the front canopy latch protruding outwardly from the bottom frame element, and the front canopy latch having a shape of a rectangular cuboid; wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, the first mount including hooked protrusions extending at either end of the first mount and a first rotating latch rotatably coupled to the first mount between the hooked protrusions, the hooked protrusions of the first mount and the first rotating latch are configured to engage the first side support of the carrier support of the stroller.

4. The assembly of claim 3, wherein the front canopy latch is engageable to secure the front of the top to the front of the bottom in a front closed position, and being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier.

5. The assembly of claim 4, wherein the back of the top is connectable to the back of the bottom in a back closed position, and is releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier.

6. The assembly of claim 3, wherein the first and second pet carrier engagements are engaged with the carrier support.
 7. The assembly of claim 6 further comprising a lower carrier support on the pet stroller, the lower carrier support comprising a front frame connected to the front lower portions, and a rear frame connected to the rear lower portions.
 8. The assembly of claim 3 further comprising a handle connected to the pet carrier.
 9. The assembly of claim 7, wherein the lower carrier support supports the bottom frame element of the pet carrier.
 10. The assembly of claim 7 further comprising a storage area located beneath the lower carrier support, the storage area including a flexible bottom and two side panels, one located on each of the first and second sides of the stroller frame.
 11. The assembly of claim 3, wherein the carrier support frame includes a continuous perimeter.
 12. The assembly of claim 3, wherein the carrier support frame is oriented in a singular flat plane.
 13. The assembly of claim 3, wherein the second pet carrier engagement comprises a second mount fixed to the bottom section, the second mount including hooked protrusions extending at either end of the second mount and a second rotating latch rotatably coupled to the second mount between the hooked protrusions, the hooked protrusions of the second mount and the second rotating latch are configured to engage the second side support of the carrier support of the stroller.
 14. The assembly of claim 3, wherein the rectangular cuboid shape of the front canopy latch includes rounded corners.
 15. The assembly of claim 3, wherein the top section further comprises a top front canopy latch on the front, the top front canopy latch protruding outwardly from the first canopy frame element, and the top front canopy latch having a shape of a rectangular cuboid.
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